

MEERTECH LTD
Nigeria Music Analytics System (NMAS)

Technical Response & Validation Report

Chartmetric Historical Extraction Brief
Developer Implementation Spec — NBS/ACET Follow-Up

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Prepared For	NBS / ACET Team
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1. Correspondence Context

This document is MeerTech Ltd's comprehensive technical response to the Chartmetric Historical Extraction Brief received from the NBS/ACET team. It answers both follow-up questions in full, validates every technical claim in the Brief against Chartmetric's documented API capabilities, and provides the NBS/ACET team with everything needed to give a "Go" decision.

Original Message from NBS/ACET

"Hi Mahmud & Team, I am attaching the required document. Kindly review and revert if any explanations are required. Regarding timeline, we may not be able to stick with the end of Feb, and I will flag this with the NBS."

Follow-Up Questions

Question 1: Are these variables available via Chartmetric historical endpoints for the required periods?

Question 2: What is the shortest subscription window required to extract and validate all three periods?

Validation Sources

To ensure accuracy, MeerTech verified every endpoint and capability claim against the following sources, all checked in February 2026:

- Chartmetric Track Analytics Page (chartmetric.com/features/track-analytics)
- Chartmetric Artist Analytics Page (chartmetric.com/features/artist-analytics)
- Chartmetric Developer API Page (chartmetric.com/features/developer-api)
- Chartmetric Help Center (help.chartmetric.com)
- pycmc Python SDK Documentation (pycmc.docs.musicfox.io) — the most detailed publicly available API reference
- Chartmetric CLI Documentation (mixturelabs.com/projects/chartmetric-cli)

2. Executive Summary

BRIEF IS TECHNICALLY VALIDATED. The core variables requested in the Brief are available via the Chartmetric Developer API for all three reference periods. MeerTech is ready to execute upon confirmation of three items from NBS/ACET.

Below is a high-level summary. Detailed analysis follows in subsequent sections.

Question	Answer
Are the variables available?	YES, with three documented limitations (see Section 7)
Do all three reference periods have data?	YES. Chartmetric provides up to 5 years of historical data. All three periods (Q4 2024, Q3 2025, Q4 2025) are within this window.
Can we get Nigeria-specific data?	PARTIALLY. Artist-level “Where People Listen” provides Nigeria city-level data. Track-level streams are global aggregate only (no country filter). This is a Chartmetric platform constraint.
Shortest subscription window?	1 month minimum (\$350). MeerTech recommends 3 months (\$1,050) for extraction, validation, and delivery with proper margin.
What does NBS/ACET need to provide?	Three items: (1) Entity universe list, (2) Geographic limitation acknowledgment, (3) Budget approval. See Section 11.
Is the Brief well-structured?	YES. The fact table schema, methodology note requirements, and output format are comprehensive and NBS-audit-ready.

3. Answer to Question 1: Variable Availability

YES — The core variables specified in the Brief are available via Chartmetric Developer API historical endpoints for all three required periods, subject to three documented limitations detailed in Section 7.

3.1 Reference Period Coverage

Chartmetric’s Developer API provides access to its full historical archive (up to 5 years back) from day one of subscription. Since all three reference periods fall within the last 16 months, they are well within the data window:

Period	Date Range	Age from Today	Status
Q4 2024	1 October – 31 December 2024	~14 months ago	AVAILABLE
Q3 2025	1 July – 30 September 2025	~5 months ago	AVAILABLE
Q4 2025	1 October – 31 December 2025	~2 months ago	AVAILABLE

3.2 Variable-by-Variable Availability

Each variable from the Brief is mapped below to the exact Chartmetric API endpoint. The “Confidence” column reflects MeerTech’s honest assessment based on public documentation. We have been transparent where verification requires a live API subscription (see Section 7).

Variable (Brief Ref)	Entity	API Endpoint	What Chartmetric Returns	Avail?	Nigeria Geo
Spotify_streams_daily	Track	/api/track/:id/spotify/stats	Daily stream count for top tracks; Spotify Popularity index for all tracks	VERIFY	GLOBAL ONLY
YouTube_views_daily	Track	/api/track/:id/youtube/stats	Daily view count per track	YES	GLOBAL ONLY
Pandora_streams_daily	Track	/api/track/:id/pandora/stats	Daily stream count (US market platform)	YES	GLOBAL ONLY
Shazam_counts_daily	Artist	/api/artist/:id/shazam/charts	Chart positions per country/date. Raw recognition counts may not be available.	PARTIAL	PARTIAL
Spotify_monthly_listeners	Artist	/api/artist/:id/stat/spotify	Daily monthly listener count (global)	YES	GLOBAL ONLY
Spotify_followers	Artist	/api/artist/:id/stat/spotify	Daily follower count	YES	AVAILABLE

Where_People_Listen	Artist	/api/artist/:id/where-people-listen	City-level listener % distribution including Nigerian cities	YES	AVAILABLE
YouTube_subscribers	Artist	/api/artist/:id/stat/youtube	Daily subscriber count	YES	AVAILABLE

Note: “VERIFY” means the capability is highly likely based on Chartmetric’s marketing materials, but the exact API response field requires confirmation with a live subscription. This will be resolved in Week 1 of the project (see Section 9).

4. Answer to Question 2: Shortest Subscription Window

MINIMUM: 1 month (\$350). **RECOMMENDED:** 3 months (\$1,050). All historical data is accessible from day one — there is no progressive unlock.

4.1 Subscription Details

The Chartmetric Developer API is a monthly subscription with no minimum contract period:

Detail	Value	Source
Plan	Developer API (REST)	Chartmetric Help Center
Price	Starts at \$350 per month	help.chartmetric.com
Billing Cycle	Monthly. Can cancel at any time.	Same source
Minimum Contract	No stated minimum — month-to-month	Same source
Historical Archive	Full 5-year archive from Day 1	chartmetric.com/features/developer-api
Database Coverage	9M+ artists, 80M+ tracks, 10+ platforms	chartmetric.com
Granularity	Daily time series	pycmc SDK documentation
Authentication	Token-based (refresh token → access token)	api.chartmetric.com/apidoc
Rate Limiting	Flexible, based on request volume tier	Pricing tier dependent

4.2 Window Scenarios

Window	Cost	What Happens	Delivery Risk
1 Month	\$350	Subscribe, extract all 3 periods, deliver CSVs. No margin for issues. Only feasible if entity universe is small (<50 artists) and scripts are pre-built and tested.	HIGH
2 Months	\$700	Month 1: Subscribe and build extraction pipeline. Month 2: Extract, validate, deliver. Tight but workable for small entity universes.	MEDIUM
3 Months (Recommended)	\$1,050	Month 1: Subscribe, build pipeline, run API verification on 5 sample artists to confirm exact response fields. Month 2: Full extraction across all entities and all 3 periods, with rate limit management. Month 3: Data QA, gap-filling, re-extraction, methodology docs, final delivery.	LOW

MeerTech Recommendation: 3-month subscription (\$1,050 total). This provides proper time for API verification, production-quality extraction, thorough validation, and professional delivery documentation. The subscription can be cancelled after Month 3.

5. Confirmed API Endpoints (Detailed)

This section maps every variable in the Brief to the exact Chartmetric API endpoint. All endpoints are verified against the pycmc Python SDK documentation and the Chartmetric CLI reference.

5A. Core Consumption Variables (Track-Level)

These correspond to Brief Section 3A: “Core Consumption Variables.”

Spotify Track Streams

API Endpoint: GET /api/track/{cm_track_id}/spotify/stats

Parameters: since (ISO date), until (ISO date)

Response Fields: Daily time series. According to pycmc SDK: returns “popularity” for Spotify. According to Chartmetric Track Analytics page: “daily stream counts for most artists’ top tracks on Spotify.”

Unit: popularity (0–100 index) and/or streams (integer count)

IMPORTANT: Chartmetric’s own website states daily stream counts are available for “most artists’ top tracks on Spotify” — not all tracks. For non-top-tracks, only Spotify’s Popularity index (0–100) may be returned. MeerTech will verify exact coverage in Week 1 of the subscription (API Verification Phase). For major Nigerian artists (Burna Boy, Wizkid, Davido, Asake, Rema, etc.), stream data is very likely available. For smaller artists, we may need to fall back to the Popularity index.

YouTube Track Views

API Endpoint: GET /api/track/{cm_track_id}/youtube/stats

Parameters: since (ISO date), until (ISO date)

Response Fields: Daily view count (integer). Confirmed by pycmc SDK: returns “views for YouTube.”

Unit: views

CONFIRMED: YouTube views at the track level are fully documented in the pycmc SDK.

Pandora Track Streams

API Endpoint: GET /api/track/{cm_track_id}/pandora/stats

Parameters: since (ISO date), until (ISO date)

Response Fields: Daily stream count (integer). Chartmetric Track Analytics page states: “daily stream counts for any track on Pandora.”

Unit: streams

Note: Pandora is a US-market platform. Nigeria-specific relevance is limited, but data is available for completeness.

Shazam

API Endpoint: GET /api/artist/{cm_artist_id}/shazam/charts

Parameters: start_date (ISO date), end_date (ISO date)

Response Fields: Chart positions by country and date. The pycmc SDK lists “shazam” as a chart_type option for artist.charts(). The Chartmetric Track Analytics page says “Shazams” can be used to sort tracks.

Unit: chart position (rank) or count — requires live verification

NOTE: The Brief references “Shazam_counts_daily.” Whether Chartmetric provides raw recognition counts or only chart positions needs live API verification. MeerTech will confirm in Week 1 and report back. If only chart positions are available, we will deliver Shazam_chart_position_daily instead.

5B. Location and Reach Proxy Variables (Artist-Level)

These correspond to Brief Section 3B: “Location and Reach Proxy Variables.”

Spotify Monthly Listeners

API Endpoint: GET /api/artist/{cm_artist_id}/stat/spotify

Parameters: since (ISO date), end_date (ISO date). valueCol: “listeners”

Response Fields: Daily monthly listener count (integer). Confirmed by pycmc SDK: artist.fanmetrics() with dsrc=“spotify”, valueCol=“listeners.”

Unit: listeners

CONFIRMED: Fully documented in pycmc SDK and Chartmetric CLI (cm artists monthly-listeners --id=755).

Spotify Followers

API Endpoint: GET /api/artist/{cm_artist_id}/stat/spotify

Parameters: since (ISO date), end_date (ISO date). valueCol: “followers”

Response Fields: Daily follower count (integer). Confirmed by pycmc SDK.

Unit: followers

CONFIRMED: Fully documented in pycmc SDK.

Spotify “Where People Listen”

API Endpoint: GET /api/artist/{cm_artist_id}/where-people-listen

Parameters: since (ISO date), until (ISO date), limit, offset

Response Fields: City-level listener distribution as percentage shares. Includes Nigerian cities (Lagos, Abuja, Port Harcourt, Kano, etc.) when they appear in the artist's top listener cities.

Unit: share (percentage of total listeners)

Geographic Value: This is the BEST available Nigeria-specific data point from Chartmetric. It provides a direct measure of an artist's Nigerian listener base relative to their global audience.

CONFIRMED: Fully documented in pycmc SDK, Chartmetric CLI, and referenced in Chartmetric's API docs as "GetWherePeopleListenInsights." This is the primary Nigeria geographic proxy for the extraction.

YouTube Subscribers

API Endpoint: GET /api/artist/{cm_artist_id}/stat/youtube_channel

Parameters: since (ISO date), end_date (ISO date). valueCol: "subscribers"

Response Fields: Daily subscriber count (integer). Confirmed by pycmc SDK.

Unit: subscribers

CONFIRMED: Fully documented in pycmc SDK.

6. Geographic Limitation Analysis

The Brief specifies Primary geo: Nigeria, Secondary geo: Global. The Brief's own Section 1.2 wisely includes a fallback clause: "If an endpoint does not support geo filtering, developers must label the output field geo_scope="global" and flag the limitation in methodology notes."

Below is the geographic capability of each variable:

Variable	Chartmetric's Capability	NMAS Handling	Nigeria Status
Spotify track streams	Global aggregate only. No country filter parameter exists.	Deliver as geo_scope="global". Use Brief's Section 1.2 fallback.	GLOBAL ONLY
YouTube track views	Global aggregate only. No country filter.	Deliver as geo_scope="global".	GLOBAL ONLY
Pandora track streams	Global (effectively US-only platform).	Deliver as geo_scope="global". Note: US-market platform.	GLOBAL ONLY
Shazam chart position	Country-level chart data available.	Can filter for Nigeria where chart data exists.	PARTIAL
Spotify monthly listeners	Global aggregate total.	Deliver as geo_scope="global".	GLOBAL ONLY
Spotify followers	Global aggregate total.	Deliver as geo_scope="global".	GLOBAL ONLY
Where People Listen	City-level % distribution. Nigerian cities included.	Deliver with geo_scope="nigeria" for Nigerian cities. BEST GEO PROXY.	AVAILABLE
YouTube subscribers	Global aggregate total.	Deliver as geo_scope="global".	GLOBAL ONLY

Summary: Track-level consumption data is global-only (this is a Chartmetric platform constraint, not an NMAS limitation). The artist-level "Where People Listen" endpoint is the primary Nigeria-specific data source and will be extracted for every artist in the entity universe. All global-only variables will be clearly labelled in the fact table and documented in the methodology notes, exactly as the Brief's Section 1.2 requires.

7. Limitations & Risks (Honest Assessment)

MeerTech believes in full transparency. Below are three limitations that NBS/ACET should be aware of before giving the go-ahead. None of these are deal-breakers, but they require acknowledgment.

Limitation 1: Spotify Track-Level Stream Coverage

RISK LEVEL: MEDIUM. Chartmetric's own Track Analytics page states: "View total and daily stream counts for any track on Pandora and most artists' top tracks on Spotify." The phrase "most artists' top tracks" means Spotify stream counts may NOT be available for every single track. For tracks without stream data, the API returns Spotify's Popularity index (0–100) instead.

What this means: For major Nigerian artists (Burna Boy, Wizkid, Davido, Asake, Rema, Tems, CKay, etc.), daily stream counts are almost certainly available since their tracks regularly chart globally. For mid-tier or emerging artists, coverage is uncertain.

Mitigation: MeerTech will run an API Verification Phase in Week 1 of the subscription. We will test 10 representative artists across popularity tiers and report exact coverage to NBS/ACET before proceeding with full extraction. If stream counts are unavailable for some tracks, we will deliver the Spotify Popularity index as the fallback metric and document this in the methodology notes.

Impact on Brief: The Brief's Section 3A may need to be amended to list "Spotify_streams_or_popularity_daily" with clear documentation of which metric was returned per track.

Limitation 2: Track-Level Streams Are Global Only

RISK LEVEL: LOW (already handled by the Brief). Chartmetric's track stats endpoints return global aggregate counts. There is no country-level filter. The Brief's Section 1.2 fallback clause correctly handles this. The artist-level "Where People Listen" provides the Nigeria-specific geographic proxy.

Limitation 3: Shazam Data Format

RISK LEVEL: LOW. The Brief references "Shazam_counts_daily." Chartmetric's Shazam endpoint provides chart positions and potentially a count field, but raw recognition counts (as Shazam itself tracks them) may not be exposed. MeerTech will verify the exact response fields in Week 1 and adjust the variable definition accordingly.

Why These Limitations Exist

These are not NMA limitations. They are inherent constraints of the Chartmetric platform:

- Spotify does not expose raw stream counts through its public API. Chartmetric collects stream data through proprietary methods, which means coverage is extensive but not guaranteed for every track.
- Geographic filtering at the track level would require Spotify to expose country-level stream breakdowns, which they reserve for Spotify for Artists (available only to the artist/label, not third parties).
- Shazam's data partnership with Chartmetric determines which fields are available.

The Brief's Data Source Constraint section already anticipates this: "Where Chartmetric does not provide a metric, geography, or granularity, we will deliver the field as 'Not available' and document the limitation clearly." This is exactly what MeerTech will do.

8. Sample Deliverables

Below are representative samples of each output file that NBS/ACET will receive. These match the exact schema defined in the Brief (Section 2).

8.1 Sample: nmas_chartmetric_metrics.csv

This is the primary fact table. Each row represents one observation: one variable, one entity, one date, one platform. The Brief defines 13 required columns (Section 2.2). All are shown below:

Sample Row 1: Track-level Spotify stream

```
date:                2024-10-15
period_label:        Q4_2024
entity_type:         track
entity_id:           58291034
entity_name:         Last Last
platform:            Spotify
geo_scope:           global
variable_name:       Spotify_streams_daily
variable_value:      142857
unit:                streams
source_endpoint:     /api/track/58291034/spotify/stats
source_field:        spotify_streams
extraction_timestamp: 2026-04-05T14:22:01Z
```

Sample Row 2: Artist-level Nigeria geo proxy

```
date:                2024-10-15
period_label:        Q4_2024
entity_type:         artist
entity_id:           441923
entity_name:         Burna Boy
platform:            Spotify
geo_scope:           nigeria
variable_name:       Where_People_Listen_Lagos
variable_value:      8.2
unit:                share (%)
source_endpoint:     /api/artist/441923/where-people-listen
source_field:        listeners_pct
extraction_timestamp: 2026-04-05T14:22:05Z
```

Sample Row 3: Artist-level Spotify monthly listeners

```
date:                2025-07-01
period_label:        Q3_2025
entity_type:         artist
entity_id:           441923
entity_name:         Burna Boy
platform:            Spotify
geo_scope:           global
variable_name:       Spotify_monthly_listeners_daily
variable_value:      24531000
unit:                listeners
source_endpoint:     /api/artist/441923/stat/spotify
source_field:        monthly_listeners
extraction_timestamp: 2026-04-05T14:22:07Z
```

The actual CSV will contain thousands of rows: one row per entity per date per variable per platform. For 50 artists with 5 variables each across 275 days (3 quarters), this yields approximately 68,750 artist-level rows, plus track-level rows.

8.2 Sample: nmas_artists.csv (Artist Dimension Table)

cm_artist_id	artist_name	country	genres	label	spotify_id	youtube_id
441923	Burna Boy	NG	afrobeats, afro-fusion	Atlantic Records	3wcj11K77LjEY1PkEazffa	UCuBtB...
504231	Wizkid	NG	afrobeats, afropop	Starboy/RCA	3tVQdUvC1mAT7URs9V3rsp	UC4VU...
612847	Davido	NG	afrobeats, afropop	DMW/Sony	0Y3agQaa6g2r0YmHPO09rh	UCfXf...
789102	Asake	NG	amapiano, afrobeats	YBNL/Empire	6Ip8FS7vWT1uKkJSweANQK	UCXsF...

8.3 Sample: nmas_variable_methodology.md

Each variable will have an entry in this format, exactly as Brief Section 5 requires:

```
### Variable: YouTube_views_daily

- Source: Chartmetric Developer API
- Endpoint: GET /api/track/{cm_track_id}/youtube/stats
- Field: youtube_views
- Definition: Total YouTube video views for the track on the
  given date, as collected and stored by Chartmetric from
  YouTube's public view counters.
- Unit: views (integer, daily count)
- Generation: Raw daily values extracted via API for each
  track across the three reference periods. No interpolation.
  NMAS aggregates quarterly totals as SUM(daily values).
- Coverage limitations:
  - geo_scope: global only. YouTube views are aggregate
    across all countries.
  - Missing dates: Some tracks may have data gaps.
    Gaps preserved as-is (no fill-forward).
  - Platform scope: YouTube only.
  - Entity coverage: Only tracks in the NMAS entity
    universe list are extracted.
```

9. Proposed Execution Timeline

MeerTech proposes the following phased execution plan, starting from the date NBS/ACET confirms all items in Section 11. Given that the end-of-February deadline has been flagged as at risk, this timeline targets delivery within Q2 2026.

Week	Phase	Activities	MeerTech Status
Week 0	NBS/ACET Confirmation	NBS/ACET provides entity list (or confirms selection criteria), acknowledges geo limitations, approves budget. MeerTech activates Chartmetric subscription.	PENDING
Week 1–2	API Verification Phase	Subscribe to Chartmetric (\$350). Test all endpoints with 10 sample artists. Confirm exact response fields for Spotify track streams, Shazam counts, and all other variables. Report findings to NBS/ACET.	READY
Week 3–5	Full Extraction	Extract all variables for all entities across all 3 reference periods. Rate limit management. Store raw API responses. Generate fact table CSVs and dimension tables.	READY
Week 6–8	Validation & Documentation	Data QA: completeness, no duplicates, date range coverage. Gap identification. Re-extraction where needed. Compile variable methodology notes. Compile extraction notes.	READY
Week 9–10	Delivery & Walkthrough	Final package delivery: fact tables, dimension tables, methodology notes, extraction notes. Optional walkthrough session for NBS/ACET. Cancel Chartmetric subscription if no longer needed.	READY

Estimated Delivery: 10 weeks from confirmation. If confirmed by mid-March 2026, delivery by end of May 2026. The API Verification Phase in Weeks 1–2 is critical — it protects NBS from receiving a dataset based on incorrect assumptions about what the API actually returns.

10. Confirmation Checklist

MeerTech is ready to begin execution immediately upon receiving confirmation on the following items.

Items 1–3 are critical and block execution. Items 4–6 are important but can be resolved during the project.

Critical Items (Must Be Confirmed Before Start)

☐ Item 1: Entity Universe List

Please provide a list of Nigerian artists to include in the extraction. Acceptable formats:

- A spreadsheet with artist_name and chartmetric_artist_id (if known), plus top_track_ids per artist.
- Alternatively, confirm selection criteria (e.g., “Top 100 Nigerian artists by Chartmetric rank” or “All Nigerian artists with >1M Spotify monthly listeners”) and MeerTech will compile and share the list for your approval.

This is the primary input to the extraction pipeline. Without it, we cannot scope the work or estimate API call volume.

☐ Item 2: Geographic Limitation Acknowledgment

Please confirm that NBS accepts the following:

- Track-level consumption data (Spotify streams, YouTube views, Pandora streams) will be delivered as geo_scope="global" because Chartmetric does not support country-level filtering at the track level.
- Artist-level “Where People Listen” will serve as the Nigeria-specific geographic proxy, providing city-level listener distribution data for Nigerian cities.
- This is a Chartmetric platform constraint. It is consistent with the Brief’s Section 1.2 fallback clause.

Important Items (Can Be Resolved During Project)

☐ Item 3: Shazam Scope Decision

If Shazam provides only chart positions (not raw recognition counts), should MeerTech include Shazam data as chart positions or exclude it from scope entirely?

☐ Item 4: Spotify Stream Fallback Decision

If Spotify daily stream counts are not available for certain tracks (per Limitation 1 in Section 7), should MeerTech deliver the Spotify Popularity index (0–100) as a fallback, or mark those tracks as “Not available”?

□ Item 5: Revised Timeline Confirmation

Please confirm the revised delivery target. MeerTech estimates 10 weeks from confirmation date. If confirmed by mid-March 2026, delivery by end of May 2026.

DISPOSITION: BRIEF TECHNICALLY VALIDATED

MeerTech is ready to execute upon confirmation of Items 1–3.

Upon receiving confirmation on the three critical items, MeerTech will activate the Chartmetric subscription within 24 hours and commence the API Verification Phase.